TAX ONE Support Dev

Plataforma de Automacao Inteligente para Suporte Fiscal

Multi-Agent AI Orchestration Platform • 11 Agentes • 10 Pipelines • 9 Slash Commands • 16.000+ Arquivos

1 Visao Geral do Projeto

O que e o TAX ONE Support Dev?

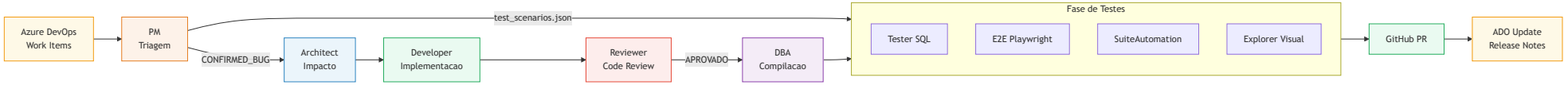
Uma **plataforma de orquestracao multi-agente** que automatiza o ciclo completo de suporte ao sistema fiscal **TAX ONE** (Thomson Reuters / Mastersaf). Desde a triagem de bugs no Azure DevOps ate a criacao de Pull Requests no GitHub, passando por analise de impacto, implementacao, revisao de codigo, testes automatizados e planejamento de sprints.

O sistema atua sobre um ERP fiscal com **4.028 tabelas Oracle**, **66.067 colunas**, **5.425 procedures/packages PL/SQL** e cobre todos os modulos fiscais brasileiros (ICMS, IPI, PIS/COFINS, ISS, SPED, eSocial, REINF, ECD, etc).

11 Agentes Especializados	**10** Pipelines Automatizados	**16.4K** Arquivos no Knowledge Base	**148 MB** Documentacao Embutida
4.028 Tabelas Oracle	**5.425** PL/SQL Procedures	**514** Specs E2E Playwright	**27+** Solucoes Documentadas

2 Arquitetura Tecnica

Fluxo Macro — Da Demanda ao Deploy



Principios de Design

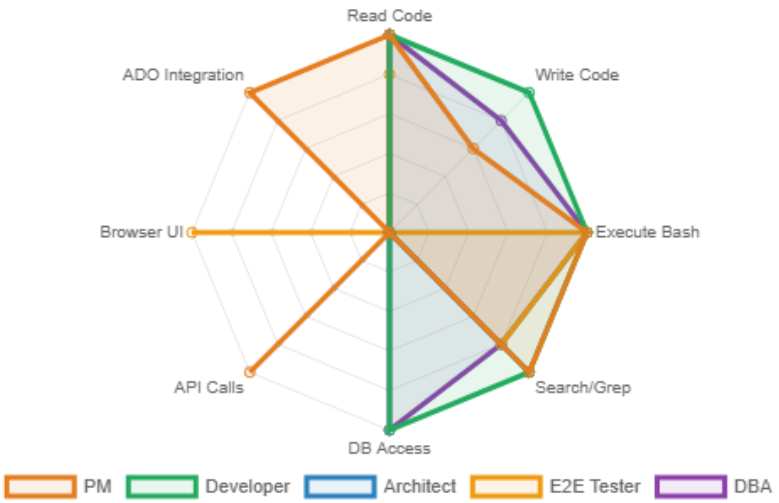
- **Knowledge-First:** Todo agente carrega contexto do KB antes de agir
- **Separation of Concerns:** Agentes com escopos definidos (read-only vs read-write)
- **Backwards-Compatible:** Fallback gracioso quando artefatos nao existem
- **Human-in-the-Loop:** Confirmacao obrigatoria em acoes destrutivas
- **Compound Engineering:** Toda solucao gera conhecimento reutilizavel

Estrutura de Diretorios

```
taxone-support-dev/
├── .claude/agents/    # 11 agentes especializados
├── skills/            # 10 pipelines automatizados
├── commands/          # 7 slash commands
├── knowledge/         # 16.4K arquivos (148 MB)
│   ├── schema/        # 4.028 tabelas documentadas
│   ├── business-rules/# 4.521 regras de negocio
│   ├── ado-fixes/     # 7.048 solucoes historicas
│   ├── webhelp/       # 3.498 artigos funcionais
│   ├── zendesk/       # 1.237 padroes de suporte
│   └── solutions/     # Compound Engineering
├── scripts/           # 10 scripts Python
└── tests/             # Evidencias locais (por WI)
```

3 Ecossistema de Agentes (11)

Mapa de Agentes por Capacidade



taxone-pm
Product Manager / Triagem

Read Write Grep Bash

Classifica WIs (bug/not-bug/feature/duplicata). Gera test_scenarios.json. Consulta FAQ, Zendesk, WebHelp.

taxone-architect
Arquiteto de Software (Read-Only)

Read Grep Glob

Analise de impacto, design proposals, mapeamento downstream, enriquecimento de cenarios.

taxone-developer
Dev Senior PL/SQL / PB / Java

Read Write Edit Bash

Implementacao de correcoes e features. PL/SQL (90%), PowerBuilder, Java/Jasper.

taxone-reviewer
Code Reviewer (Read-Only)

Read Grep Glob

Seguranca, qualidade, performance, convencoes. Threshold de confianca 75%+.

taxone-dba
Oracle DBA Specialist

Read Write Edit Bash

Explain plans, indexes, DDL review, compilacao PL/SQL no banco (porta 1521).

taxone-angular
Angular 11 Developer

Read Write Edit Bash

Frontend: Angular 11, RxJS, Material, Wijmo FlexGrid. Monorepo taxsami_taxone_frontend.

taxone-tester
Test Script Expert

Read Write Bash

taxone-e2e-tester
Playwright E2E Tester

Read Bash Grep

taxone-suite-runner
SuiteAutomation Regression

Read Bash Grep

Scripts SQL de validacao, test data setup, roteiros manuais, regressao.

514 specs em 7 modulos. Automacao via playwright_runner.py.

Framework Java, compara expected vs obtained (ISO-8859-1). Oracle local.

taxone-sm

Scrum Master Automator

Read | Bash | Grep

Sprint planning, Story Points, alocacao Dev/Tester por especialidade.

taxone-explorer

Visual Exploratory Tester

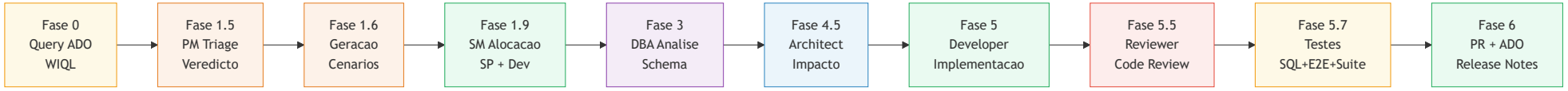
Bash | Read | Write

Teste exploratorio via browser. Screenshots, navegacao guiada, validacao visual.

4 Pipelines Automatizados (10 Skills)

Pipeline Principal: /taxone-auto-fix

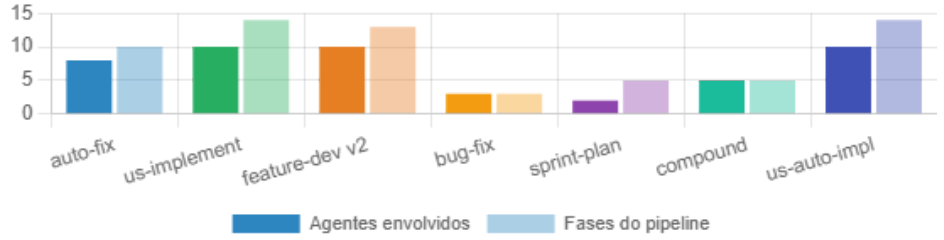
Fluxo completo: busca WIs no ADO, triagem, implementacao, testes, PR e atualizacao do ADO.



Skills Disponveis

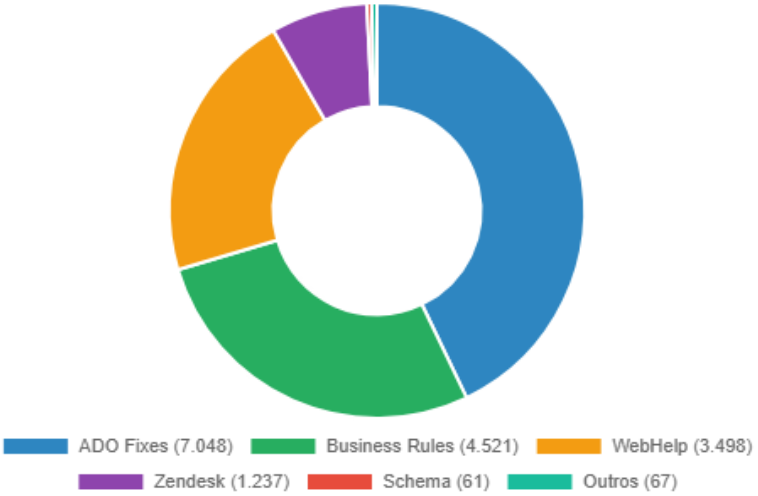
Skill	Proposito	Autonomia
/taxone-auto-fix	Bug fix completo ADO→PR	Total
/taxone-us-auto-implement	User Story implementacao	Total
/taxone-bug-fix	Bug fix manual (leve)	Semi
/taxone-sprint-plan	Planejamento de Sprint	Total
/taxone-feature-dev	Dev generico: 13 fases, 10 agentes, 4 camadas teste	Total
/taxone-us-implement	Feature/US com regras + design	Total
/taxone-compound	Documentar solucao (KB)	Total
/taxone-doc-gen	Geracao de documentacao	Total
/query-executor-prod-uat	SQL em UAT/Prod de clientes	Manual

Fases e Agentes por Pipeline

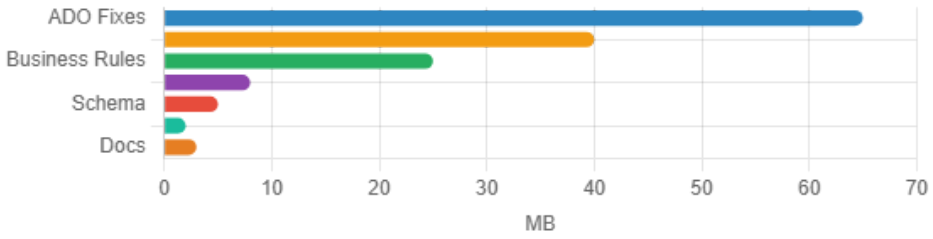


5 Knowledge Base — 16.432 Arquivos

Distribuicao por Categoria



Tamanho por Categoria (MB)



ADO Fixes (7.048)

Historico completo de Work Items resolvidas com metadados, PR cache e vertical/modulo mapeados. Inclui angular, estadual, federal, evolucao tecnica.

Business Rules (4.521)

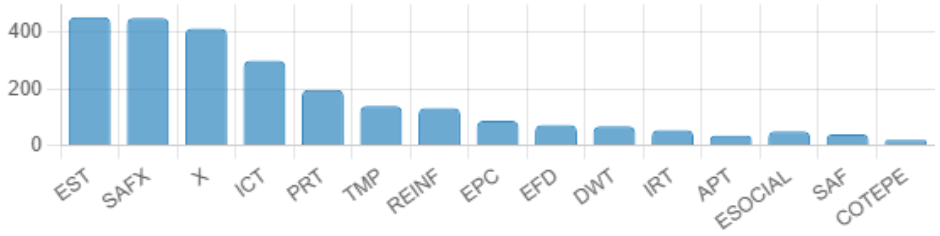
Regras de negocio por vertical: Estadual, Federal, Municipal, SPED, Basicos, Especificos, Report Fiscal. 66 subdiretorios cobrindo todo o sistema.

WebHelp (3.498)

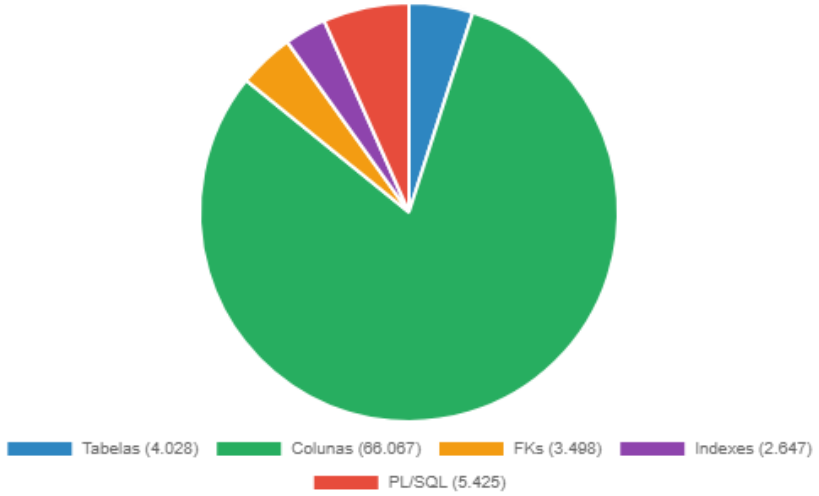
Artigos funcionais extraidos do OneSource WebHelp: FAQs, manuais, interfaces, patches, migration. Indexado por MODULE_INDEX.json.

6 Schema Oracle — 4.028 Tabelas

Tabelas por Prefixo (Top 15)



Cobertura do Schema

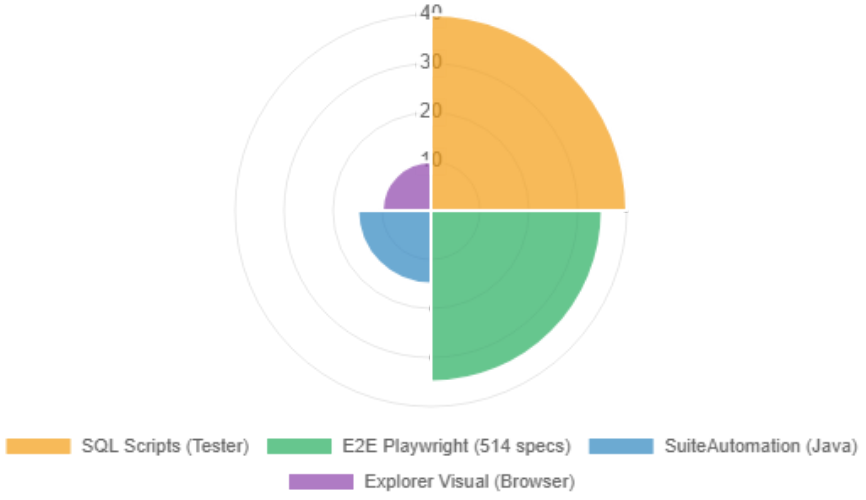


Artefatos do Schema Knowledge

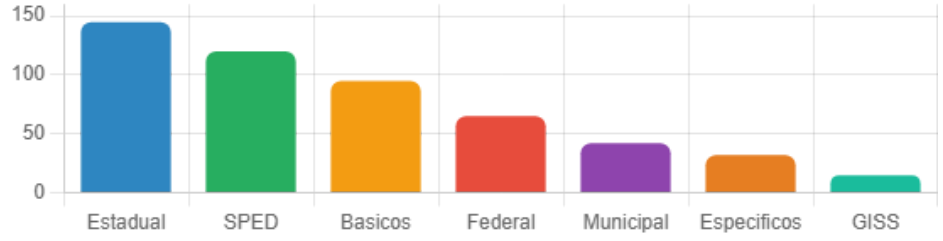
Artefato	Conteudo	Uso
61 arquivos .md	4.028 tabelas, 66.067 colunas, 3.498 FKs, 2.647 indexes	Referencia para todos os agentes
COLUMN_GLOSSARY.md	380 colunas recorrentes com dominio	PM/Architect para entender campos
PLSQL_MAP.md	344 views + 5.425 procedures → tabelas	Architect para impacto, Developer para navegacao
MODULE_MAP.md	91 prefixos DB → modulos funcionais	PM para classificacao, SM para alocacao
TABLE_HOTSPOTS.md	Tabelas/procs mais citados em bugs	PM para prioridade, DBA para proatividade
RELATIONSHIPS.md	3.498 FKs cross-module com diagrama	Architect para downstream analysis

7 Infraestrutura de Testes

4 Camadas de Teste



Specs E2E por Modulo



Test Scenarios JSON — Gerado na Triagem (Fase 1.6)

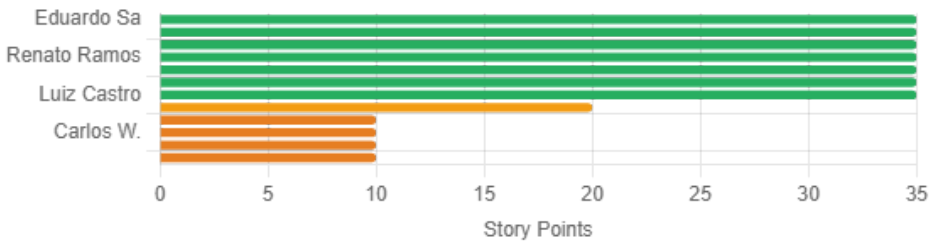
O PM gera cenarios estruturados que alimentam todas as camadas de teste automaticamente.

happy_path	negative	edge_case	regression
Fluxo principal funciona	Erros sao tratados	Limites e excecoes	Nada quebrou

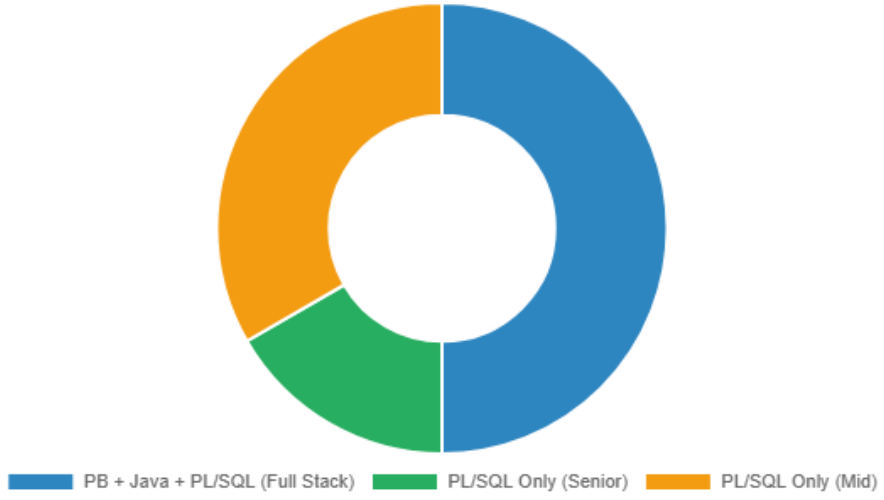
```
tests/{WI_ID}/
├── test_scenarios.json         ← PM gera (Fase 1.6)
├── 01_setup_test_data.sql      ← Tester expande sql_hint
├── 02_validacao_cenario_*.sql  ← Tester cria por cenario
├── e2e_report.txt              ← E2E usa e2e_grep do cenario
├── regression_report.txt       ← SuiteAutomation compara
├── exploratory_report.md       ← Explorer segue steps[]
└── explore_*.png               ← Screenshots da validacao
```

8 Sprint Planning Automatizado

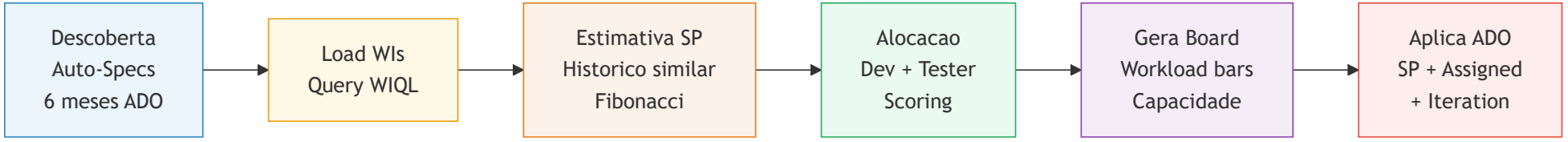
Capacidade do Time



Distribuicao de Especialidades



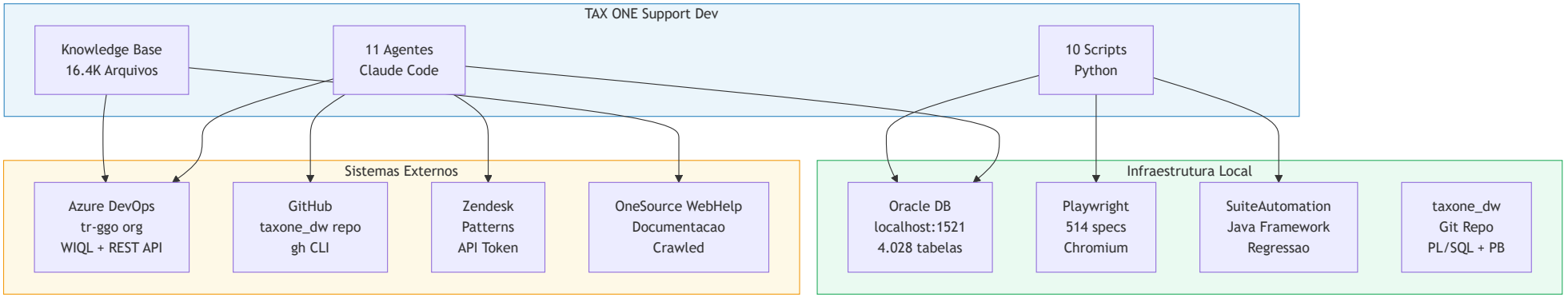
Fluxo de Sprint Planning



Scoring de alocacao: Especialidade no modulo (+30), vertical (+20), tecnologia (+15), capacidade disponivel. **12 devs** com capacidades de 10 a 35 SP/sprint. Automatico via `/taxone-sprint-plan`.

9 Integracoes Externas

Ecossistema de Integracoes



Azure DevOps

- WIQL queries (batch)
- Work Item CRUD (PATCH)
- Attachments upload
- Discussion comments (HTML)
- Tags automaticas (4 tipos)
- Release Notes template

GitHub

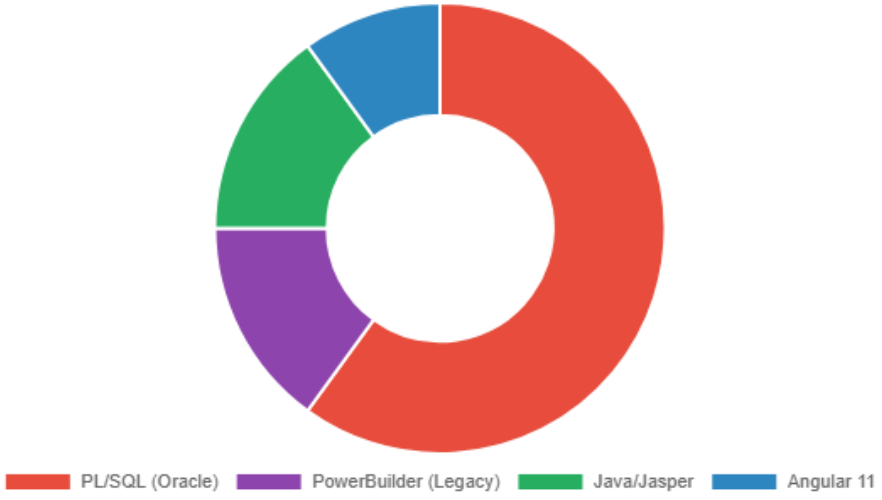
- Branch: MFS{WI_ID}
- Commit: [DEV] MFS{ID} - desc
- PR create via gh CLI
- Link ADO: AB#{WI_ID}
- Code review automatico
- ISO-8859-1 binary edit

Oracle Database

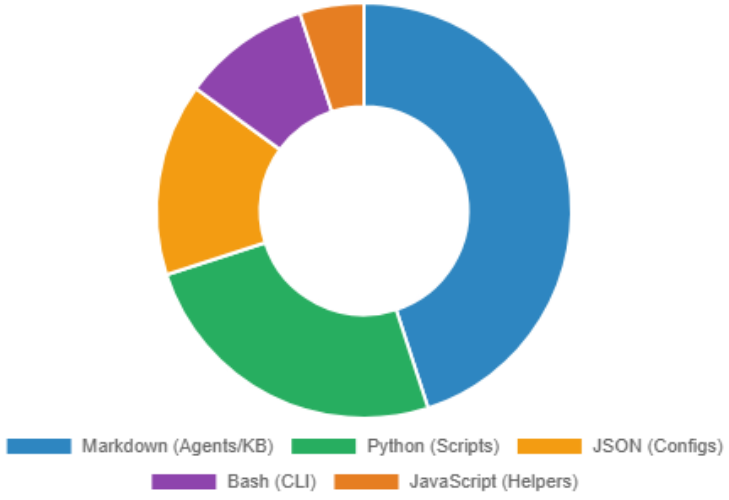
- Compilacao PL/SQL (DBA)
- Schema extraction (thin)
- Test data setup/cleanup
- SuiteAutomation compare
- Porta 1521 (somente)
- V2R010AC@MFSPDB

10 Technology Stack

TAX ONE (Sistema Alvo)



Plataforma (Este Projeto)



Regras de Operacao Criticas

Regras de Seguranca

- PL/SQL: ISO-8859-1 + CRLF (binary edit, nunca Edit tool)
- DDL: NUNCA alterar existente (criar nova com WI_ID)
- Compilar PL/SQL ANTES de testes (porta 1521)
- tests/ NUNCA vai para Git (somente ADO Attachments)
- Confirmar antes de: branch, commit, push, PR, ADO update

Tags Automaticas no ADO

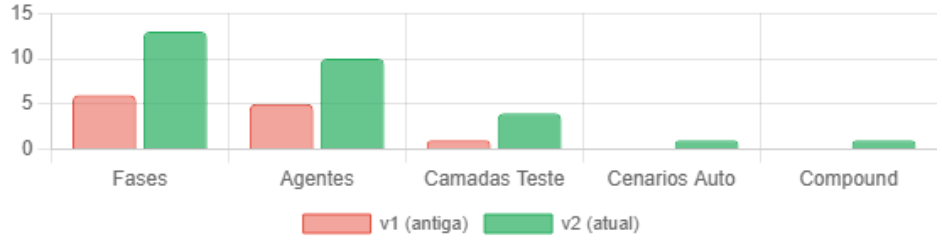
- SUST-IA-CLAUDE Em toda WI atualizada
- SUST-IA-CLAUDE-NOT-BUG WI nao e bug
- SUST-IA-CLAUDE-ANALISE Precisa analise extra
- SUST-IA-CLAUDE-REGRA Envolve regra de negocio

11 Comparacao dos Pipelines

Matriz de Funcionalidades por Pipeline

Funcionalidade	auto-fix	us-implement	us-auto-impl	feature-dev v2	bug-fix
Integracao ADO	Sim	Sim	Sim	Nao	Nao
PM Triagem	Sim	Sim	Sim	Manual	Nao
test_scenarios.json	Sim	Sim	Sim	Sim	Nao
Architect (impacto)	Sim	Sim	Sim	Sim	Nao
DBA + Compilacao	Sim	Sim	Sim	Sim	Sim
Angular (frontend)	Sim	Sim	Sim	Sim	Nao
Tester SQL	Sim	Sim	Sim	Sim	Nao
SuiteAutomation	Sim	Sim	Sim	Sim	Nao
E2E Playwright	Sim	Sim	Sim	Sim	Nao
Explorer Visual	Sim	Sim	Sim	Sim	Nao
Cobertura de Regras	Nao	Sim	Sim	Sim	Nao
Compound Engineering	Sim	Sim	Sim	Sim	Nao
Sprint Planning (SM)	Sim	Nao	Sim	Nao	Nao
Release Notes	Sim	Sim	Sim	Nao	Nao
Total de Fases	10	14	14	13	3
Agentes Utilizados	8	10	10	10	3

Quando usar cada pipeline?

Cenario	Pipeline
Bug com WI no ADO	/taxone-auto-fix
Feature/US com WI no ADO	/taxone-us-auto-implement
Feature/US com regras formais	/taxone-us-implement
Dev generico sem ADO	/taxone-feature-dev
Bug fix rapido sem ADO	/taxone-bug-fix
Planejamento de Sprint	/taxone-sprint-plan
Documentar solucao no KB	/taxone-compound
Gerar documentacao tecnica	/taxone-doc-gen
SQL em UAT/Prod do cliente	/query-executor-prod-uat

Evolucao feature-dev: v1 → v2